The Dissociation Problem in Hartree–Fock Theory: Restricted vs. Unrestricted Potential Energy Curves of H_2 and LiH

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1 Introduction

The Hartree–Fock (HF) method approximates the many-electron wavefunction by a single Slater determinant built from molecular orbitals expanded in a finite basis set [Szabo and Ostlund, 1996].

In its restricted form (RHF), α and β electrons share the same spatial orbitals, which gives a well-behaved description of closed-shell molecules near their equilibrium geometry. However, RHF is known to fail when a bond is stretched: the closed-shell determinant of H_2, $|\sigma_g\alpha\,\sigma_g\beta|$, contains equal weights of covalent (H· ·H) and ionic (H^+H^-) configurations at every internuclear distance, so it cannot dissociate into two neutral atoms [Coulson and Fischer, 1949, Szabo and Ostlund, 1996].

The unrestricted formalism (UHF) removes this constraint by allowing α and β electrons to occupy different spatial orbitals. Beyond a certain distance, known as the Coulson–Fischer point [Coulson and Fischer, 1949], a lower-energy broken-symmetry UHF solution appears in which each electron localises on a different atom.

In this work we compute the RHF and UHF dissociation curves of H_2 and LiH at the HF/3-21G level and compare them with the energy of the separated atoms, in order to characterise the dissociation problem of RHF and how UHF overcomes it.

2 Methods

Every calculation was done with Gaussian 09 (Rev. D.01) [Frisch et al., 2013] at the HF level with the split-valence 3-21G basis set [Binkley et al., 1980]. Molecules were defined with a Z-matrix in which interatom distance r is a variable, and potential energy curves were obtained with the `Scan` keyword.

Since both molecules are diatomic, r is the only internal coordinate. The H_2 scan ran from 10.0 to 0.5 Å in steps of 0.25 Å (39 points), and the LiH scan fro 7.5 to 1.2 Å in steps of 0.3 Å (22 points). Both scans start at the dissociated geometry, so tahteach point usesthe converged wavefunction of the previousone as its initial guess.

Each molecule was computed with two methods: RHF, and UHF with the `guess=mix` keyword, which mixes the HOMO and LUMO in the initial guess to break the spatial symmetry between α and β orbitals. Without this mixing, UHF converges to the restricted solution for a closed-shell singlet. The reference energies of the separated atoms, E(H) and E(Li), were obtained from UHF/3-21G calculations on the doublet ground states. The dissociation curves are reported as

$$\Delta E(r) = E_{\mathrm{AB}}(r) - E(\mathrm{A}) - E(\mathrm{B}), \tag{1}$$

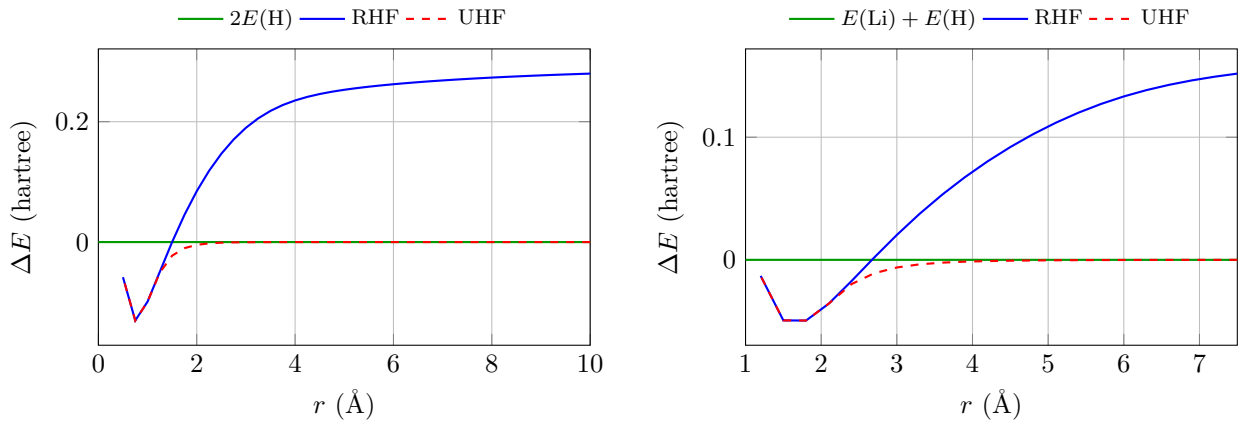
so that a correct curve must tend to $\Delta E = 0$ as $r \to \infty$.

The workflow was automated with Bash scripts on a Linux System. A loop launched every input file through Gaussian and skipped those already terminated normally, and a second one read the atomic energies from the output files (`grep`), extracted the scan summary tables and computed ΔE for every point, writting one data file per curve.

The figures were generated directly from these files with PGFPplots/Ti*k*Z [Feuersänger, 2012].

3 Results

Figure 1 shows the dissociation curves and Table 1 summarises the main quantities.



(a) H_2: $\Delta E = E(H_2) - 2E(H)$ (b) LiH: $\Delta E = E(LiH) - E(Li) - E(H)$

Figure 1: RHF and UHF (`guess=mix`) dissociation curves of H_2 and LiH at the HF/3-21G level. The green line marks the energy of the separated atoms.

Both panels share the same structure. The blue (RHF) and dashed red (UHF) curves overlap around the minimum, where the restricted closed-shell solution is the most stable one, and separate at intermediate distances. Beyond that point, only the UHF curve approaches the green line, i.e. the energy of the separated atoms.

Table 1: Equilibrium distance r_e, dissociation energy D_e, Coulson–Fischer point r_{CF} and RHF error at the largest distance computed, HF/3-21G. Experimental values from [Huber and Herzberg, 1979].

	r_e (Å)		D_e (eV)		r_{CF} (Å)	$\Delta E_{RHF}(r_{max})$ (eV)
	HF	Exp.	HF	Exp.		
H_2	0.735	0.741	3.55	4.75	≈ 1.2	7.62
LiH	1.64	1.596	1.42	2.52	≈ 2.2	4.13

Equilibrium region. At short distances the RHF and UHF curves coincide exactly, because the restricted solution is the lowest one: both methods give the same minimum. HF/3-21G reproduces the equilibrium distances within 0.05 Å, but underestimates the dissociation energies, by 25 % for H_2 and 44 % for LiH. This is the expected behaviour of Hartree–Fock, which lacks electron correlation and therefore underbinds.

Dissociation region. Beyond the minimum the two methods diverge qualitatively. The RHF curves never approach the separated-atom limit: at the largest distance computed, they lie 0.280 hartree (H_2, 10 Å) and 0.152 hartree (LiH, 7.5 Å) above it, errors larger than the bond energies themselves. In contrast, the UHF curves separate from RHF at the Coulson–Fischer point (≈ 1.2 Å for H_2 and ≈ 2.2 Å for LiH) and converge smoothly to $\Delta E = 0$, as they should. The Coulson–Fischer point lies further out for LiH because its bond is longer, so the restricted solution remains stable up to larger distances.

The improvement of UHF comes at a cost: the broken-symmetry wavefunction is no longer an eigenfunction of $\hat{S}^2$. At large separation $\langle \hat{S}^2 \rangle \to 1$, instead of the value 0 of a pure singlet: the UHF solution is an equal mixture of singlet and triplet states (spin contamination).

4 Conclusion

- Near equilibrium, RHF and UHF are identical and give reasonable bond lengths, but underestimated dissociation energies due to the lack of electron correlation.
- RHF cannot describe bond dissociation: its closed-shell determinant keeps a constant 50 % ionic contribution, so the energy of the separated fragments is too high by several eV.
- UHF with a symmetry-broken initial guess dissociates correctly into neutral atoms beyond the Coulson–Fischer point, at the price of spin contamination ($\langle \hat{S}^2 \rangle \to 1$).
- The UHF solution depends on the initial guess: without `guess=mix` it collapses onto the RHF solution. A balanced description of the whole curve requires methods that go beyond a single determinant, such as configuration interaction.

References

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